CS482 Software Engineering Project: Black Jack

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1 Client Information

By sharing this client information and the rest of this document, you are stating that this client has provided this project as something they want (not something you created and asked if they wanted), and that they are interested in having you complete this project for your capstone.

- Client name: Dr. Nguyen Ho
- Client title: Assistant Professor
- Client email address: tnho@loyola.edu
- Client employer: Loyola University of Maryland
- How you know the client: Professor at our university

2 Project Description

2.1 Overview

Our client requested a browser-based blackjack application that generates revenue through advertisements. Players will use virtual money and may participate as guest or as registered players. The application will include a lobby with multiple tables, different buy-ins, human and bot participants, and social features. The interface will use a green theme with a company name and logo developed by our team. Advertisements will appear in the lobby and at game tables, preferably in side panels. The project will also include administrative tools for managing users, distributing virtual money, responding to feedback, and viewing application analytics.

2.2 Key Features

- Accounts and guest access: Registration, login, and temporary guest participation with assigned or selected display names.
- Virtual money: A registration bonus, displayed balances, and administrator-issued bonuses for individual users, selected groups, or all users.
- Lobby and tables: A list of active tables showing buy-ins, available seats, and game status.
- Table configuration: Table creation, optional password protection, invitations, bot selection, and dealer assignment.
- Blackjack gameplay: Human and bot participation, a single deck, and reshuffling according to the rule clarified with the client.

- Social features: Friend lists, invitations, friend messages, table chat, and private messages between table participants.
- Player profiles: Avatars, balances, the most recent 5–10 games, and access to additional game history.
- Tutorial: A skippable video tutorial that users can replay.
- Feedback: A mailbox through which users can contact administrators and receive responses.
- Account management: User management, virtual-money adjustments, usage statistics, and application performance information.
- Visual presentation: A green theme, original branding, and optional animations and sound effects.

2.3 Why this Project is Interesting

This project gives an opportunity to combine an interactive game with the technical challenges of a real-time, multiuser application. Players see consistent game information, take turns correctly, and receive accurate updates to their virtual balances. The project also goes beyond gameplay into authentication, database design, social communication, administration, and analytics. These components make it suitable for a capstone because they require several areas of computer science to work together.

2.4 Areas of CS required

- Software engineering: Design, incremental development, testing,
- Web development/ Application: Web Interface, lobby navigation, game controls, profiles, and administration screens.
- Databases: Storing Account information, balances, game history, messages
- Cybersecurity: Passwords, input validation, and cheating prevention
- Data analysis: Summarizing player activity and monitoring application performance.

2.5 Potential Concerns and Questions

The main technical concern is maintaining a smooth web gameplay when multiple users act, disconnect, or leave a table. The server must control game rules and balances so that players cannot cheat in the game through browsers. The requested feature set is also big. We are not sure if the combination of gameplay, bots, messaging, advertising, and analytics is achievable within the capstone timeline and which features should receive priority.

- Virtual money: What are the starting bonuses, allowed buy-ins, and funding rules for guests?
- Advertisements: Will the capstone use demonstration advertisements or a live advertising integration?
- Analytics: Which player statistics and performance measurements does the client want?
- User management: Should removing an account delete its records or deactivate access while preserving game history?

2.6 Summary of Efforts to Find a Project

Team met with Dr. Ho to discuss her request for a software project. During the meeting, we gathered requirements for gameplay, account access, visual design, social features, administration, and analytics. Dr. Ho also asked the team to propose features and develop the application's name and logo.

3 Requirements

3.1 Non-Functional Requirements

Table 3 presents the NFRs for this software project.

ID	NFR Title	Category	Description
NFR1	Color Scheme	Usability	Color scheme should be green
NFR2	Game response	Performance	Game should respond fast to the users choice, response between game and user should be no more than 50 ms
NFR3	Animations and sounds	Usability	Add animations that run smoothly on around 60fps, music and sound should have a latency less than 100 ms and should not cut out
NFR4	Permissions	Security	Add permissions to different roles so admins can access certain pages user cannot
NFR5	Deck reshuffle	Usability	Reshuffle the deck after each round
NFR6	Protection of accounts	Security	Protect the users account and data by encrypting the information

Table 1: Non-Functional requirements

3.2 Functional Requirements (User Stories)

Table 2 presents the functional requirements for this software project.

ID	Story Title	Points	Description
A1	Add/Remove Users	5	As an admin, I want to add or remove users so that I can manage who can play the game
A2	Add Money	1	As an admin, I want to add money to the users so that they can keep playing the game.
A3	Data analysis	3	As an admin, I would like to get data of people using the app, so that I can keep an analysis of the performance of the app.
A4	Ad revenue	2	As an admin, I want the game to have several ads so that we can generate revenue out of the game.
A5	Feedback chat	1	As an admin I would like to have a feedback chat, so users can give me their feedback.
A6	Web Design	1	As an admin I would have green theme and a game logo.
A7	Tutorial video	2	As a user, I want to have a tutorial to watch, so that I can understand the rules and know how to play.
P1	Log In	1	As a player, I want to create an account, so that I will keep my data of the game.
P2	Exit/ReJoin	2	As a player, I want to create an account, so that I will keep my data of the game.
P3	Switch Tables	2	As a player, I want to switch game tables, so that I can pick my prefered game table.
P4	Money Counter	1	As a player, I want to see how much money I currently have, so that I can know how many games I can play.
G1	Guest Log In	1	As a guest, I want to skip the sign up, so that I dont have to log in.
F1	Customize Profile	1	As a user, I want to write a username, so that I can customize my username.
U1	Share The Game	3	As a user, I want to share the game, so that I can play with friends.

T1	table accessibility	1	As a user, I want to make my the table public or private, so that I can only invite my friends.
U2	Player Selection	1	As a user, I want to choose players, so that I can play with real people or bots.
CH1	Chat with Friends	5	As a player, I want to have a chat box, so that I can send messages to my friends in the game.
CH2	chat with individual at table	1	As a player, I want to have a chat box at the table, so that I can send messages to specific players at the table.
U1	Table Security	1	As a user, I want to set a password for a table, so that I can keep it private.
U4	Manage Tutorial	1	As a user, I want to be able to skip, watch or rewatch the tutorial.
F3	Friends List	2	As a player, I want to have a friend list, so that I know who i am friends with.
F4	Player History	3	As a player, I want to have the history of all the games I played, so that I can see my scored of all the games.
U5	Table Status	2	As a player, I want to see the status of current tables, so that I can see how many tables are available.
U6	Submit a ticket	1	As a user, I want to submit a ticket to admin, so that i can send feedback or problems about the game.
U7	Start the game	2	As a user, I want to press "start" the blackjack card game
U8	Game simulator	1	As a user, I want the game to be functioning and follow the game rules of Black Jack.
U9	Game Buttons	1	As a user, I want to stand, hit, double, split during the game
U10	Betting	1	As a user i want to have initial betting and keep betting after.
U11	Split option	1	As a user I want to have two hands for an option to split.
U12	End the game	1	As a user I want to end the game when i want to.
	Total:	87	

Table 2: Functional requirements as User Stories.

3.3 AI Analysis on Functional Requirements

Source: OpenAI GPT-6 Astr Prompt: Act as senior Software Engineer to help you find any missing or hidden requirements. Look closer to the highlighted sections and give me good feedback Reflection: A lot of our user stories lacked detailed explanation on some user stories. We also saw many repetitions across the stories where we could have taken a different approach and extended in broader to get specific functions.

4 System Design

4.1 Architecture

We are going to use a 3 tier architecture level dividing our work in the following tiers:

- Presentation Layer: This will include things the users can see, such as the designs of the cards, chips, table and User interface.
- Application layer: This will include the things that the user does not need to see but are essential for the game such as starting a game, getting cards adding friend and all other u functions that the game needs.
- Data layer: This layer will store data such as the user information that will be needed for their accounts.

4.2 Diagrams

[CS482, on sprints/iterations 2-3, you need to create and update a diagram (check Iteration 2-3 assignment for which type of diagram). On CS496, since before sprint/iteration 1, you should have a class diagram and keep it up-to-date. In CS496, if your class diagram changes at each sprint, then create a Class Diagram subsection for the sprints, and show the changes; while keeping the one here the most up-to-date version.]

4.3 Technology

For programming languages we are going to use HTML, CSS, JavaScript, for the databases we are going to use MongoDB to store our information. For the Frameworks we are planning to use react native as our main framework, Node.js and for web development we are going to use AWS amplify and lastly we using version control GitHub.

4.4 Coding Standards

For our coding standards, we like to commit only code with unit tests with a 80% or above coverage. For naming tables and functions we are going to use lowercase and if they include more than one word we are separating with a "_" for example shuffle_cards as a function. When committing a code we are demanding our peers to write a very well written description of what they are committing. We also want to add comments inside the code in order to describe what that portion of the code does so other people can know what we are working on. Before committing code, a person must make sure that it was well tested and it works without breaking other people's code so that we do not run into a problem when committing.

4.5 Data

Collection	**fields**
User	id, username, email, password, role, balance, friends, avatar, gameWins
Table	gameID, password, tableName, isPublic, status, maxPlayers, activePlayers
game History	id, userID, tableID, betAmount, hand, dealerHand, outcome, payout
chat	Id, senderID, recieverID, message, time

Table 3: Data

4.6 UI Mocks

Figma prototype: View the full UI design in Figma

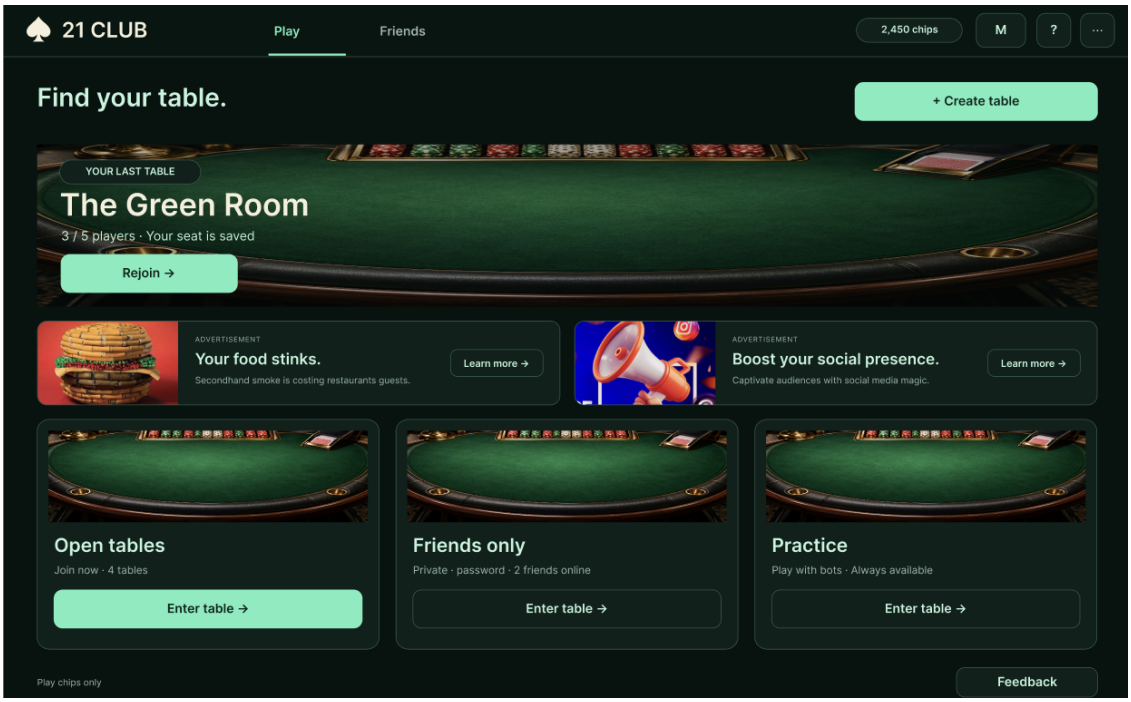


Figure 1: The main page welcomes users to the Blackjack application and provides navigation to start a game, rejoin the game, practice and connect with friends.

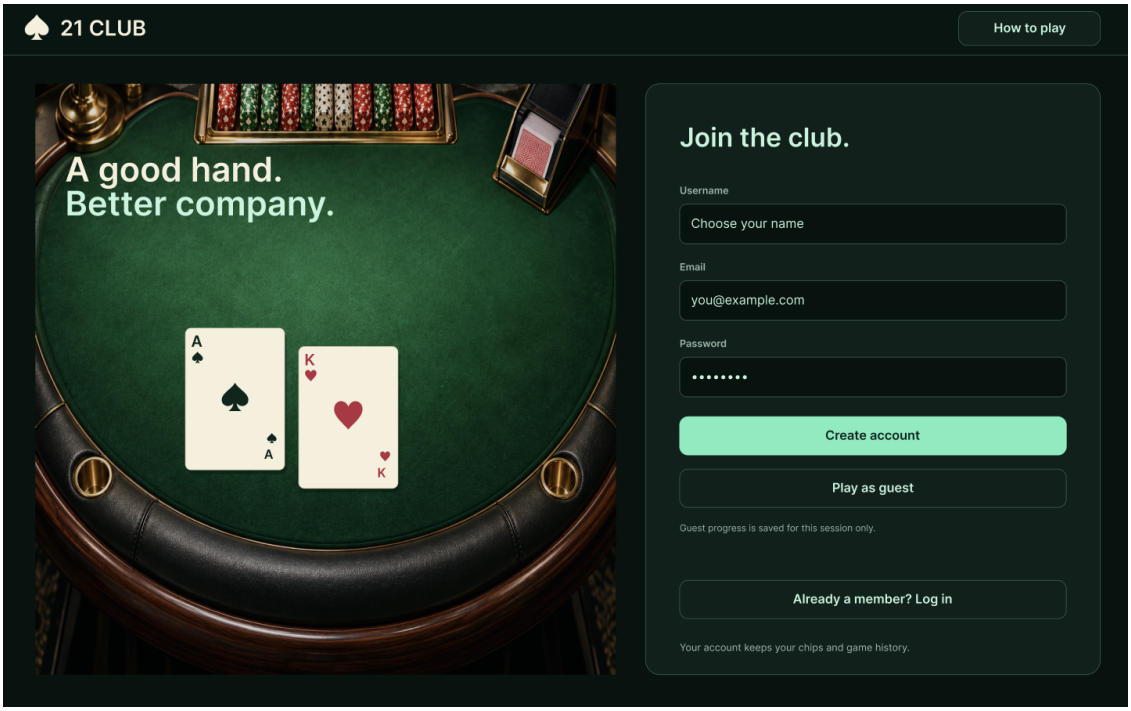


Figure 2: The login page allows registered users to access their accounts by entering their username or email and password. It displays an error message if the credentials are incorrect, but users can play as guest as well.

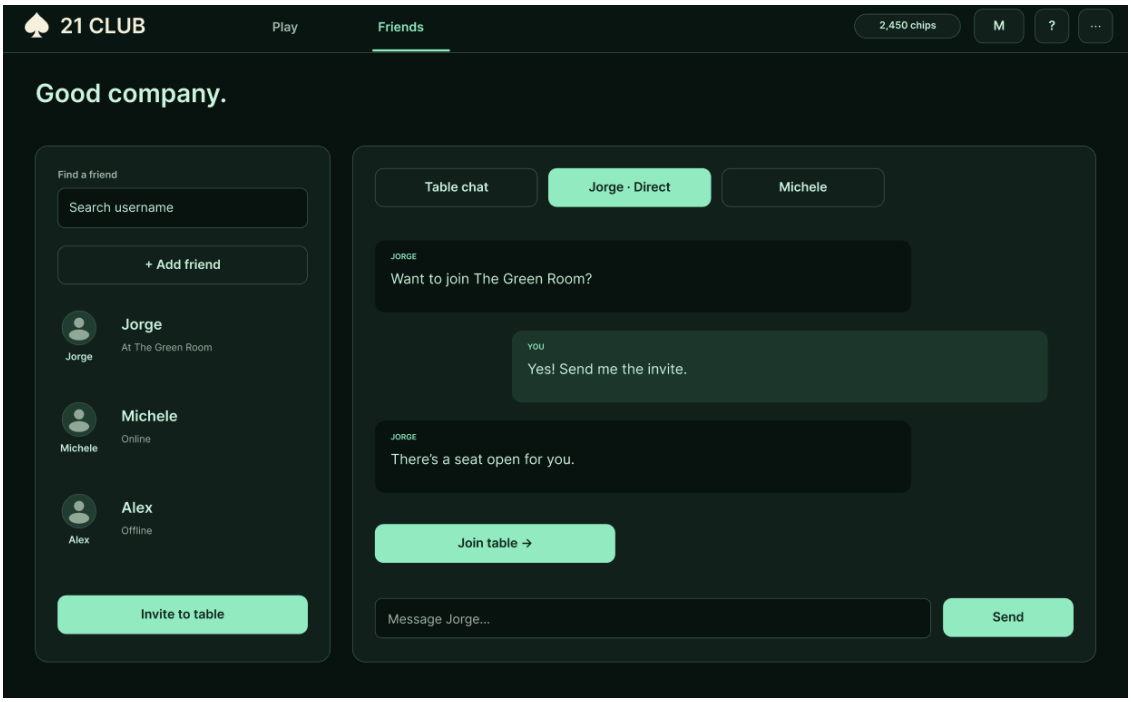


Figure 3: The chatbox allows users to send and receive messages with friends. It displays the conversation history, identifies message senders, and provides a text field and send button for new messages..

5 Iterations

5.1 Iteration Planning

[In CS496, you plan all iterations beforehand. In CS482, you update the planning here at each iteration.]

Table 4 shows the iteration planning.

			Points	
It.	**Dates**	**Stories**	**Planned**	**Done**
1	10/12 - 10/15	P4 "Money Counter, P1 "Log In", G1 "Guest Log In" , F4 "Player History" , U7 "Start the game", A7 "Game Tutorial", U8 "Game Simulator", U9 "Game Buttons", U10 "Betting", U11 "Split Option" , U12 "End the game"	15	15
2	02/01 - 03/01	S3 Story Title, S4 Story Title, S5 Story Title, S6 Story Title	17	15
3	03/01 - 04/01	S7 Story Title, S8 Story Title, S9 Story Title, S10 Story Title, S11 Story Title	21	20
4	04/01 - 05/01	S12 Story Title, S13 Story Title, S14 Story Title, S15 Story Title	19	19
5	05/01 - 06/01	S16 Story Title, S17 Story Title	06	06
		Total:	70	65

Table 4: Iteration Planning for Incremental Deliveries

5.2 Iteration/Sprint 1

5.2.1 Planning

For Sprint 1, we are planning to get the game simulator to run as a single user with no fancy features integrated. The first iteration will cover User Stories:

- P4 "Money Counter" 1 point -Jorge
- P1 "Log In" 1 point -Michele
- G1 "Guest Log In" 1 point -Michele
- F4 "Player History" 3 point -Michele
- U7 "Start the game" 1 point -Mushtariy
- A7 "Game Tutorial" 2 point -Mushtariy
- U8 "Game Simulator" 2 point -Mushtariy/Jorge *SPIKE: finish by October 16th
- U9 "Game Buttons" 1 point - Jorge
- U10 "Betting" 1 point - Jorge
- U11 "Split Option" 1 point - Jorge
- U12 "End the game" 1 point- Jorge

The total points for this iteration will be 15. The reason behind the planning is to get the game started. The initial start should be a game simulator to have an actual initial base where we can build it off from and make it more complex that fits the client needs. The major features we will be focusing on: building the working game with one user, to give the user an option to sign up as a player or continue the game as a player, and to keep the player's game history.

5.2.2 Work Done

The stories completed so Far:

- P4 "Money Counter" 1 point -Jorge *1 hour*
- P1 "Log In" 1 point -Michele * 1 hour *
- G1 "Guest Log In" 1 point -Michele * 1 hour *
- F4 "Player History" 3 point -Michele *2 hours*
- U7 "Start the game" 1 point -Mushtariy *0.5 hour*
- A7 "Game Tutorial" 2 point -Mushtariy *2 hours*
- U8 "Game Simulator" 2 point -Mushtariy *2 hours*
- U9 "Game Buttons" 1 point - Jorge *2 hours*
- U10 "Betting" 1 point - Jorge *2 hours*
- U11 "Split Option" 1 point - Jorge *1 hour*
- U12 "End the game" 1 point- Jorge *1 hour*

5.2.3 Testing Coverage

The testing covers 93.61% line coverage with no failed tests so it is relatively a good coverage because it hits a majority of the lines and passes them which shows that the code in functioning the way it supposed to. To increase coverage we can write more tests on uncovered lines like 93-98 and 149-156. Duplicated lines were decreasing the line coverage and causing issues in the code for the game to run. As suggested in class we will use NiCAD or iClones to prevent this issue for Iteration 2.

Figure 4 shows the test coverage

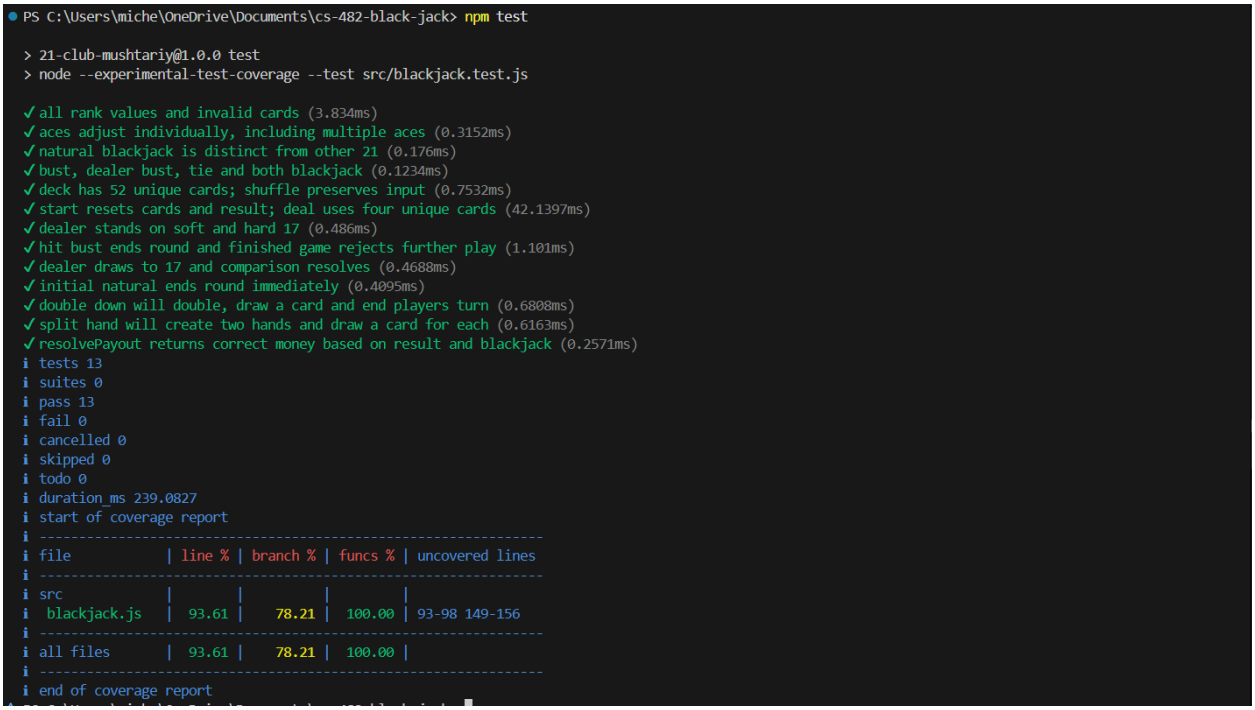


Figure 4: Iteration 1 test coverage report

5.2.4 Retroespective & Reflection

Michele — Backend: Michele worked on connecting the application to MongoDB and setting up the API server. One of the main challenges was getting the frontend, backend, and database to communicate correctly so that player information and game history could be saved and retrieved.

Jorge — Building on the Existing Code: Jorge had difficulty building on Mushtariy's existing code because he first needed to understand its structure and how the components worked together. Identifying where to place navigation buttons and add new features required careful coordination to keep the interface consistent and avoid disrupting existing functionality.

Mushtariy — Game Setup and Branch Integration: Mushtariy faced challenges setting up the game logic, initializing the game state, and connecting it to the interface. Another major challenge was merging all three team members' branches into one working version. Resolving merge conflicts involved identifying duplicate code, fixing syntax errors, and making sure everyone's contributions worked together after the merge.

As a team, we worked well with timing and coordination. We hit all 15 points and it took us around 15.5 hours to complete iteration excluding the PDF and Testing. One thing we will improve on during Iteration

2 is commit messages. We will need to have direct messages stating what exact changes were made in the code to avoid merge conflict.

5.3 Iteration/Sprint 2

5.3.1 Planning

[Which stories did you plan for this iteration/sprint. Add the total points for this plan. You can also explain the reason behind your planning, and what major feature(s) your team is focusing on delivering by completing these stories. You may use a table for a summary display of the planning, but elaborate in text more detail in your focus and feature plan.]

5.3.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.3.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.3.4 Retroespective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

5.4 Iteration/Sprint 3

5.4.1 Planning

[Which stories did you plan for this iteration/sprint. Add the total points for this plan. You can also explain the reason behind your planning, and what major feature(s) your team is focusing on delivering by completing these stories. You may use a table for a summary display of the planning, but elaborate in text more detail in your focus and feature plan.]

5.4.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.4.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.4.4 Retroespective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

5.5 Iteration/Sprint 4

[CS496 has 5 sprints. CS482 only has only 3 sprints (remove Iterations 4 and 5 from this doc if you are writing a doc for 482]

5.5.1 Planning

[Which stories did you plan for this iteration/sprint. Add the total points for this plan. You can also explain the reason behind your planning, and what major feature(s) your team is focusing on delivering by completing these stories. You may use a table for a summary display of the planning, but elaborate in text more detail in your focus and feature plan.]

5.5.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.5.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.5.4 Retroespective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

5.6 Iteration/Sprint 5

5.6.1 Planning

[Which stories did you plan for this iteration/sprint. Add the total points for this plan. You can also explain the reason behind your planning, and what major feature(s) your team is focusing on delivering by completing these stories. You may use a table for a summary display of the planning, but elaborate in text more detail in your focus and feature plan.]

5.6.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.6.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.6.4 Retroespective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

6 Final Remarks

6.1 Overall Progress

[Have you completed everything? If so, present evidence on how you brought value to your client, and the overall client satisfaction. Otherwise, estimate how much progress you done and how long it would take to finish this project. Be concrete about your progress, you know how many story points your software is, how many points you completed (this shows your progress). You also how many points your team delivers at each iteration, therefore you can estimate how many more iterations it would take to finish the leftover points (show the math).]

6.2 Project Reflection

[Your personal reflection on the project. What lessons did you learned. What would you have done differently? How can you do better work in future projects? You may write this as a team or per person (or both — if all your iterations were team reflections, then it would be better to write individual reflections here)]

Appendix

[Appendix section if needed]